

FlowPay

Payment Risk and Revenue Optimization

End-to-End Fintech Analytics Case Study

Analytics and Risk Intelligence Division | 2024

Dataset: ~1 million transactions | ~25,000 users | ~3,000 merchants

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01. Executive Summary

FlowPay is a rapidly growing digital payments platform processing nearly one million transactions annually across multiple payment methods, merchants, and user segments. As the platform scaled, a set of interconnected operational and risk-related challenges emerged -- from high payment failure rates and rising fraud exposure to refund-heavy merchant categories and systemic revenue leakage.

This case study documents a structured, business-focused analytical investigation into FlowPay's payment ecosystem. The central finding was that the platform's most critical challenge was not fraud, but payment failures -- which accounted for nearly 70 percent of total revenue leakage. Retry mechanisms recovered only approximately 25 percent of failed transaction value, meaning the majority of lost revenue was permanently unrecoverable without systemic intervention.

Metric	Value
Transactions Analysed	Approximately 1 million (annual)
Total Revenue Leakage	Approximately Rs. 810 million
Projected Recoverable Revenue	Rs. 275 million -- Rs. 350 million+
Baseline Transaction Success Rate	Approximately 88%
Failure Rate	Approximately 9.4%
Fraud Rate	Approximately 2.4%

02. Business Problem

FlowPay's leadership lacked a unified, data-driven understanding of the factors driving revenue erosion and operational inefficiency. Decisions were largely reactive -- responding to individual symptoms rather than addressing root causes.

Specifically, the following questions were unanswered:

- Why are transactions failing, and which payment methods and channels are most affected?
- Which users and merchants are contributing most to fraud and refund exposure?
- How do fraud patterns evolve across different user segments and merchant categories?
- Are refund and fraud issues isolated incidents or indicators of systemic problems?
- How much revenue is the platform losing across failure, fraud, and refund channels combined?

At the scale of a platform processing millions of rupees in transactions daily, even marginal inefficiencies compound into material financial losses. The need for a systematic, evidence-based diagnostic was clear.

03. Analytical Framework

The investigation was structured into eight interconnected analytical layers, each designed to answer a specific business question while feeding into the overarching revenue and risk assessment.

L a y e r	Area	Scope
	1 Platform Health Analysis	Transaction growth, GPV trends, success, failure, fraud, and refund rates over time
	2 Payment Performance Analysis	Failure rates by payment method, device, geography, and transaction value tier
	3 Merchant Risk Analysis	Fraud and refund concentration by merchant and category; outlier detection
	4 User Behavior Analysis	Retry patterns, velocity signals, device switching, and geographic anomalies
	5 Revenue Loss Analysis	Loss quantification across failure, fraud, and refund channels; retry recovery effectiveness

6	Conversion Funnel Analysis	Attempt to success to refund mapping; leakage point identification at each stage
7	Fraud and Refund Type Analysis	Classification by fraud typology and refund reason category
8	Risk Scoring Model	Behavioral segmentation producing Low, Medium, and High risk user classifications

04. Key Findings

Finding 1: Payment Failures Drive the Majority of Revenue Loss

Payment failures were identified as the dominant source of revenue leakage, contributing approximately 70 percent of the Rs. 810 million total loss -- far exceeding fraud at 17 percent and refunds at 14 percent. This finding fundamentally reframed the platform's core risk priority: the central question shifted from how to stop fraud to how to reduce failed payments and recover lost revenue.

Loss Category	Contribution	Estimated Value
Payment Failures	~70%	~Rs. 567 million
Fraud Losses	~17%	~Rs. 138 million
Refund Losses	~14%	~Rs. 113 million
Total	100%	~Rs. 810 million

Finding 2: Retry Recovery Is Severely Underperforming

Although retry success rates appeared reasonable at the transaction count level, the value-weighted recovery picture was poor. Of Rs. 562 million in failed transaction value, retries recovered only Rs. 140 million -- a recovery rate of approximately 25 percent. The remaining Rs. 421 million was permanently unrecovered. High-value transactions showed the weakest retry success rates, amplifying the overall financial impact.

Metric	Value
Total Failed Transaction Value	Rs. 562 million
Recovered via Retry	Rs. 140 million
Permanently Unrecovered	Rs. 421 million
Value-Weighted Recovery Rate	Approximately 25%

Finding 3: Platform Stability Is Adequate But Not Optimised

FlowPay's overall transaction success rate of approximately 88 percent is within acceptable industry ranges. However, at the platform's transaction volumes, a 9.4 percent failure rate and 2.4 percent fraud rate translate into material financial consequences. A one to two percentage point improvement in first-attempt success rate alone would unlock approximately Rs. 50--80 million in recoverable annual revenue without any increase in acquisition costs.

Finding 4: Merchant Risk Is Concentrated, Not Systemic

The vast majority of FlowPay merchants operated within normal risk parameters. However, a small subset of merchants -- particularly in Gaming, Real Estate, and Electronics -- contributed disproportionately to both fraud exposure and refund volumes. This concentration pattern is operationally significant: it enables targeted, surgical interventions rather than broad platform-wide restrictions.

Category	Risk Level	Primary Concern
Gaming	High	Elevated fraud and chargeback rates
Real Estate	High	Large-value fraud concentration
Electronics	High	Refund abuse and return fraud
Travel	Medium	Moderate refund rates with seasonal spikes
Food and Beverage	Low	Low fraud; minor operational refunds

Finding 5: High-Value Users Are Lower Risk Than Assumed

Contrary to conventional risk assumptions, high-spending users demonstrated greater payment reliability, lower fraud incidence, and reduced failure rates compared to the broader user base. The platform also showed a healthy user distribution with no unhealthy dependence on a small cohort of power users -- a positive indicator for long-term revenue stability.

Finding 6: Behavioral Signals Outperform Static Risk Indicators

Static signals such as device count and geographic spread showed limited standalone predictive power for fraud detection. Dynamic behavioral indicators -- particularly transaction velocity and retry intensity -- were significantly stronger discriminators. Fraud detection accuracy improved substantially when multiple behavioral signals were combined into a composite risk score.

Signal Type	Signal	Predictive Strength
Behavioral (Dynamic)	Transaction velocity	High
Behavioral (Dynamic)	Retry intensity	High
Static	Device count	Low (standalone)
Static	Location changes	Low (standalone)
Composite	Combined score	Strong

Finding 7: Fraud Is Primarily Identity-Driven

FlowPay's fraud landscape was dominated by identity-related attack vectors rather than payment-level manipulation. Account Takeover, Identity Theft, and Device Spoofing accounted for the majority of confirmed fraud events. This pattern underscores the importance of authentication controls, session monitoring, and behavioral anomaly detection over purely transaction-level fraud filters.

Fraud Type	Description	Detection Priority
Account Takeover	Legitimate accounts accessed by unauthorised actors	Critical
Identity Theft	Synthetic or stolen identity used at signup	Critical
Device Spoofing	Device fingerprint manipulation to bypass controls	High

Finding 8: Refunds Are Mostly Operational, Not Behavioural

Unlike fraud, refunds were primarily driven by operational and processing issues -- wrong transaction amounts, duplicate charges, and merchant-side errors. This distinction is strategically important: reducing refunds requires process and systems improvements, not fraud controls.

Refund Reason	Classification	Recommended Action
Wrong Amount	Operational	Strengthen pre-authorisation amount validation
Duplicate Transaction	Operational	Implement idempotency keys at the gateway layer
Merchant Error	Operational	Merchant process audits and operational training

05. Risk Scoring Model

A behaviour-based risk scoring model was constructed to segment the user base into Low, Medium, and High risk cohorts. The model incorporated five primary behavioral signals:

- Transaction velocity: rate and volume of transactions within rolling time windows
- Retry intensity: frequency and timing of payment retry attempts
- Transaction value behaviour: deviation from established user spending baselines
- Device activity: abnormal patterns in device usage and switching frequency
- Geographic movement: anomalous location-based transaction clustering

Validation Results

Risk Segment	Fraud Rate	Revenue Contribution	Recommended Treatment
Low Risk	0%	Significant	Frictionless experience; standard monitoring
Medium Risk	~2%	Moderate	Soft step-up verification on high-value transactions
High Risk	~4%	Meaningful	Real-time friction; enhanced authentication controls

A critical business insight from the model: high-risk users generated meaningful platform revenue. Blanket blocking would suppress fraud but simultaneously destroy business value. The optimal strategy is to deploy intelligent, risk-calibrated friction rather than binary access controls.

06. Revenue Impact and Recovery Opportunity

The combined analysis quantified total identifiable revenue leakage at approximately Rs. 810 million. Of this, a substantial portion is structurally recoverable through targeted operational and system improvements -- without increasing customer acquisition spend.

Intervention Area	Estimated Recovery Potential	Time to Impact
Payment success rate improvement	Rs. 120M -- Rs. 160M	3 to 6 months
Retry system optimisation	Rs. 80M -- Rs. 100M	1 to 3 months
High-value transaction recovery	Rs. 50M -- Rs. 70M	1 to 3 months
Operational refund reduction	Rs. 25M -- Rs. 40M	3 to 6 months

Total Projected Recovery	Rs. 275M -- Rs. 370M+	--
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07. Strategic Recommendations

Recommendation 1: Improve First-Attempt Payment Success

Priority: Critical

Invest in gateway routing optimisation, card-level friction reduction, and payment infrastructure reliability. The target should be a 2 to 3 percentage point improvement in first-attempt success rate. At current transaction volumes, each percentage point improvement unlocks approximately Rs. 50-80 million in recoverable annual revenue.

Recommendation 2: Redesign the Retry Recovery System

Priority: High

Implement intelligent retry timing using optimised window detection. Prioritise high-value failed transactions for personalised recovery flows. Improve retry user experience to reduce customer abandonment during the recovery window. Target: increase the value-weighted recovery rate from the current 25 percent to 45 percent or above.

Recommendation 3: Deploy Risk-Based Authentication Controls

Priority: High

Replace binary access controls with a tiered friction model calibrated to the risk scoring output. Low-risk users receive a fully frictionless experience. Medium-risk users receive soft step-up verification on high-value transactions. High-risk users receive real-time authentication challenges before transaction processing.

Recommendation 4: Implement Targeted Merchant Risk Controls

Priority: Medium

Focus enhanced monitoring and intervention on the Gaming, Real Estate, and Electronics categories. Establish merchant-specific fraud thresholds, automated flagging workflows, and clear escalation protocols. Avoid platform-wide restrictions that would impose friction on the 90+ percent of merchants operating within normal parameters.

Recommendation 5: Reduce Operational Refund Volume

Priority: Medium

Address root causes systematically: implement idempotency keys to eliminate duplicate transactions, strengthen pre-authorisation amount validation logic, and conduct merchant-facing process audits in high-refund categories. Target a 30 to 40 percent reduction in operationally-caused refunds within six months.

08. Methodology and Technical Approach

This analysis was conducted entirely in SQL, demonstrating that sophisticated business intelligence and risk analytics can be delivered without specialised machine learning infrastructure when the analytical framework is well-structured.

Area	Detail
Dataset Scale	Approximately 1 million transaction rows, 25,000 users, and 3,000 merchants across a multi-table relational schema
SQL Techniques	Complex joins, common table expressions (CTEs), window functions, conditional aggregations, cohort and funnel analysis
Segmentation Method	Behavioral risk scoring using a weighted composite signal model across five behavioral dimensions
Funnel Analysis	Attempt to success to refund conversion mapping with leakage quantification at each stage
Revenue Modelling	Loss quantification across three channels with recovery scenario modelling and sensitivity estimates
Data Source	AI-generated synthetic dataset (Tonic.ai) designed to simulate realistic fintech transaction patterns and behaviors

Dataset Considerations

As with all synthetic datasets, certain characteristics differed from production environments. Merchant activity distribution was more uniform than real-world data, behavioral variation was modestly compressed, and fraud signal separation was somewhat weaker than typically observed on live platforms. Findings should be interpreted as directionally valid rather than as production-grade benchmarks. Despite these constraints, the dataset was highly effective for stress-testing analytical methodologies and demonstrating the end-to-end analytics workflow.

09. Conclusion

FlowPay operates a strong, scalable payments platform with a healthy user base and broad transaction participation. However, the platform is losing a material amount of revenue annually -- not primarily due to fraud, but due to addressable payment reliability issues and an inefficient recovery system.

The analysis demonstrates that the highest-impact recovery opportunity lies in three areas: improving first-attempt payment success rates, redesigning retry recovery infrastructure, and deploying behaviourally-calibrated risk controls. Together, these interventions could recover an estimated Rs. 275--350 million or more in annual revenue without increasing customer acquisition costs.

The broader analytical lesson is that the most visible problem is rarely the biggest one. At FlowPay, payment reliability eclipsed fraud as the primary driver of revenue loss. Improving existing systems can generate more value than acquiring new users -- a principle that applies equally to most platforms operating at scale.

This case study was developed as part of a practical analytics portfolio focused on SQL-based business analysis, fintech risk analytics, and decision-oriented problem solving. The objective was to simulate how real analytics teams investigate operational and financial problems inside digital payment companies -- moving from raw data to business decisions.